

The Social Consequences of Watch-Time Optimization

Abstract

Social media platforms have become major sources of news and information, but recommendation algorithms influence the content users consume and the viewpoints they encounter. This study investigates how YouTube’s watch-time optimization algorithm affects information exposure in political and health-related topics. Using a dataset of 22,193 YouTube videos focused on political and health topics, we analyzed engagement patterns, content themes, and viewpoint diversity. Our methods included K-means clustering to group videos by engagement and duration, BERTopic modeling to identify content themes, and a Topic Entropy calculation to quantify the content’s viewpoint diversity. Our findings identified three content archetypes: Low-Impact Snippet, Factual Rundown, and Rabbit Hole Perpetrator. Results showed that while Factual Rundown videos dominate watch time, long-form “rabbit hole” videos foster deep immersion. Topic modeling of the Rabbit Hole Perpetrator cluster revealed specific themes that often spark polarizing debates. Topic modeling and topic entropy analyses helped investigate the concentration of a limited number of topics, suggesting narrowed viewpoints in the content. Overall, our findings show how YouTube’s watch-time optimization creates a localized, high-duration sphere of information that prioritizes engagement over introducing users to new perspectives, reinforcing existing beliefs. This suggests that watch-time optimization contributes to narrow viewpoints through this immersive deep-dive content, reinforcing existing beliefs and certain perspectives.

Introduction

With the rise of social media platforms, many have switched from traditional sources to these platforms for their news. YouTube now has about 2.7 billion active users and is one of the most influential media and information sources. However, compared to other sources, YouTube not only distributes content but also uses an algorithm to tailor and curate it for a specific viewer. This is done using an algorithm that prioritizes watch time and engagement. This poses the question of what specific content is being shown to users.

The social consequences of these algorithms are not fully understood; however, researchers have begun to warn [6] that these systems reward more polarizing and sensational content, as such videos tend to hold engagement longer. This is especially concerning in health and politics, where misinformation can have detrimental effects on voters and patients. These can influence how people vote and how medical diagnoses are made, and misinformation can easily be mistaken for truth. This is often referred to as the “rabbit-hole” effect [2], in which viewers start with moderate content and are led deeper into increasingly extreme, narrower content as they continue viewing.

In our study, we analyze 22,193 videos across these two categories to answer the question: Does the YouTube recommendation algorithm’s preference for high watch time lead to informational concentration of more extreme or less diverse viewpoints within sensitive topics such as politics and healthcare? To conduct our research, we used K-means clustering to classify

videos by engagement and duration, BERTopic semantic modeling to identify themes within clusters, and Topic Entropy to quantify viewpoint diversity.

This paper is organized into sections, with Section 2 discussing prior work on recommendation algorithms, filter bubbles, and polarization in online spaces. Section 3 discusses our dataset and methods in more detail. Section 4 presents our key findings and results, and Section 5 explores their implications. Our paper ends with a discussion of limitations and future research possibilities in Section 6.

Related Works

In recent years, the importance of recommendation algorithms has increased as social media platforms have become a popular source of news. Previous work has delved into this topic, focusing on personalized algorithms, political radicalization, and misinformation. These studies provided insight into the broader social consequences of recommendation algorithms. In particular, it raises questions about how engagement recommendations affect viewpoint diversity and, in this case, those viewpoints pertaining to politics and healthcare.

An influential study conducted by Ribeiro et al. [1] examined whether users move from mainstream political YouTube content to more extreme opinions through the recommendation algorithm. This study analyzed over 330,000 videos and millions of user interactions on each video. From this, they found that YouTube users tend to move from less extreme communities to more intense political communities. This study provided essential research on the “rabbit hole” effect from recommendation pathways.

Although this research helped in investigating the effects of YouTube’s algorithm on the “rabbit hole” effect, Ledwich and Zaitsev [2] observed that the algorithm favored mainstream content over extremist content. Their study examined recommendation pathways across 800 political channels and found no preference for extremist content. This suggested that YouTube might have begun to curb radicalization on the platform by making changes to its algorithm. Between these two studies, it is critical to acknowledge whether recommendation algorithms push extremist content or whether there is another form of content narrowing that cannot be measured.

Other studies have focused beyond radicalization and have investigated whether the recommendation algorithm confirms pre-existing beliefs. Haroon et al. [3] conducted a study that found YouTube’s recommendations regularly aligned with the user’s political preferences, which could reduce their exposure to alternative political views. This study further amplifies concerns that engagement-driven content contributes to content filters and reinforces pre-existing views.

While the YouTube recommendation algorithm contributes to the “rabbit hole” effect, which could polarize users’ viewpoints, another major effect is the spread of misinformation. Studies have shown that misleading health information, specifically about vaccines, was mainly spread on YouTube. [4]. That said, misinformation about political issues has also been found to fuel online polarization. This was because engagement-driven platforms favor content that grabs viewers, not necessarily content based on accurate information.

The polarization of users' political views is due not only to the previously mentioned "rabbit hole" effect but also to another consequence of the recommendation algorithm. Lewis [5] explores the idea that recommendation pathways gradually narrow a user's information stream. This could result in users finding content similar to what they have been viewing rather than radical or extremist content.

These prior studies explore how YouTube recommendations lead users to extremist content, and our study asks whether the algorithm exposes people to less diverse information. Measuring the "rabbit hole" effect and informational isolation among YouTube videos is beneficial.

Data and Methods

For our dataset, we analyzed a collection of 22,193 YouTube Videos from a 1.5 GB raw file of about 600,000 videos provided by our professor. Our focus was on areas where misinformation would likely have negative effects on viewers and on their interpretation of important social issues. To filter our dataset and focus on the domains of health and politics, we utilized two categories of keywords. To filter out videos related to health, we used the following keywords: diabetes, obesity, anxiety, depression, keto, steroids, vaccine mandate, substance abuse, diet, vegan, performance enhancing, and carnivore diet. This process was repeated for the politics domain, and the words used included: abortion, gun control, voter fraud, border crisis, pro-life, Ukraine War, LGBTQ, Green New Deal, gun policy, and pro-choice. The result was videos whose transcripts included at least one key term within the video's content. This led us to the final dataset of 22,193.

Then, for each video, we collected video details including title, channel name, and publication date, along with engagement metrics: views, likes, comments, and duration. These numerical values were used for our K-means algorithm discussed in the following section. YouTube also provides a generated transcript, which we used for our content-level analysis during the BERTopic semantic modeling step described in our methods section.

We adopted a multi-step framework that combined keyword-based data collection, clustering, topic modeling, and topic entropy analysis to examine YouTube's recommendation algorithm. To begin, we developed our list of 22 keywords listed in the Data section above. Our choice of topics was based on our consideration of informational and polarizing video content. Based on our keywords, we extracted videos from our dataset, including the initial search results. For the videos we extracted, we recorded the video ID and its corresponding transcript. The videos without transcripts were removed to maintain consistency.

Our next step was to perform feature engineering using video length (seconds) and engagement metrics, which were based on views, likes, and comments. To reduce the influence of highly skewed distributions, we performed individual log transformations followed by standardization. Using these variables, we then applied K-means clustering with $k = 3$ to group videos into distinct archetypes. We strategically chose $k = 3$ because it cleanly divides the videos into 3 dominant behaviors of the YouTube video recommendation system.

The clusters were then labeled as follows: Factual Rundown, Low-Impact Snippet, and Rabbit Hole Perpetrator. Factual Rundown videos are highly engaging; they usually feature shorter, moderate-length informational content. Low-Impact Snippet videos are short-form videos that tend to have lower engagement. Rabbit Hole Perpetrator videos are longer-form with higher retention. After proper labeling, we compared the distributions of the metrics across these clusters. With this comparison, we can examine if the recommendation algorithm prioritizes videos with higher engagement over videos with informational diversity, that is, videos that focus on providing proper reports on topics instead of focusing on gaining engagement. To address our question about the algorithm's preference for watch-time optimization, we calculated the total potential watch-time volume (in hours) for each cluster. This is calculated by multiplying the view count by the duration in seconds, then dividing by 3600 seconds per hour.

To further dissect the content of the videos in our dataset, we applied BERTopic, which leverages contextual embeddings generated from a pre-trained model. We processed our transcripts by removing English stopwords before topic modeling. We also created a custom stopword list to filter out common filler words in YouTube content, such as "Uh" and "Um." Then, we applied BERTopic to cluster transcripts and extract essential keywords. To properly quantify viewpoint diversity and polarization, we calculated Topic Entropy from the distribution of BERTopic clusters.

Our final examination on our dataset was a system-level analysis of the recommendation pathways, rather than network analysis, which focuses on individual click-by-click pathways. Although our initial proposal discussed mapping recommendation networks, we decided to pivot to Topic Entropy as a polarization measure to simplify the coding process, since our BERTopic input consisted of over 22,000 videos. Topic Entropy was calculated using the topic distribution frequencies as percentages. We then calculated Shannon Entropy using the standard information-theory formula (see appendix) applied to these formulas. Overall, the entire pipeline described in this section provided key insights into engagement patterns and viewpoint diversity across our dataset.

Results

Our analysis of 22,193 YouTube videos revealed distinct structural differences in how content is consumed on the platform. By applying K-Means clustering to the videos based on engagement (views, likes, comments) and duration, we identified three content archetypes.

Cluster 0 is **Factual Rundown**, representing high-engagement, mainstream content. This cluster contains 8,060 videos and has the highest median engagement among the clusters (36,737.5 views, 1,626.5 likes, 278 comments). The median length was 631.5 seconds, a medium amount. Despite only accounting for about 36% of the 22,193 videos, this cluster dominates watch time with about 660 million total watch time hours, about 94% of the total watch time of the corpus. This signifies that viral, mainstream content draws the most public attention.

In contrast, Cluster 1, the **Low-Impact Snippet**, has the highest video count (8,668) but accounts for only about 0.07% of watch time. These videos are short, with a median length of 157 seconds, and have very low engagement metrics (370 views, 6 likes, 1 comment).

Cluster 2, the **Rabbit Hole Perpetrator**, provides the most valuable results for addressing the research question. These are the long-form, deep-dive videos. They account for about 25% of the corpus and about 5.5% of total watch time, with a moderate level of engagement (2081 views, 65 likes, 9 comments). Although these videos do not attract a large audience, especially compared to Cluster 0's 94% statistic, they have a significantly higher median duration of 1,527 seconds, over 25 minutes. This means the rabbit hole cluster captures content not intended for virality but for immersive, long-form viewer retention within that smaller audience. These metrics reveal the "rabbit hole" nature of the videos in this cluster, demonstrating that the platform's algorithm sustains this immersive, deep-dive content despite lower overall view counts, thereby optimizing watch time.

An important component of our research question is the specific content being presented within the rabbit hole. This way, we can analyze the viewpoints represented. We performed BERTopic modeling specifically on the Rabbit Hole cluster. The results showed that polarizing topics are prevalent in this cluster (see appendix). Topic 1 is a key example of this, as it centers on high-profile, often controversial figures (trump, biden, harris). Topics 3 and 6 are very religious-focused (god, lord, jesus, praise). Topics 4, 5, and 7 are focused on specific healthcare and mental health topics (mind, brain, kidney, health). Topic 9 contains legal terms, and Topic 10 pertains to LGBTQ culture. The ideological theme of Topic 8 is unclear, with representations such as black, rolling, network, etc. Topic 2 has conversational YouTuber terminology, which is not useful for our purposes.

We analyzed viewpoint diversity using a metric we defined as Topic Entropy, determined by the statistical distribution of the BERTopic clusters. Applying this metric to the Rabbit Hole cluster, we calculated a Topic Entropy score of 2.80 with an evenness index of 0.84. While this evenness index indicates that content is distributed across several sub-topics within the cluster, our topic modeling revealed that the topics encompass polarizing, non-mainstream themes. This demonstrates that while the rabbit hole has some different silos, such as partisan politics or alternative health, the overall environment isolates users from broad, mainstream factual diversity.

Total Potential Watch Time Volume by Cluster

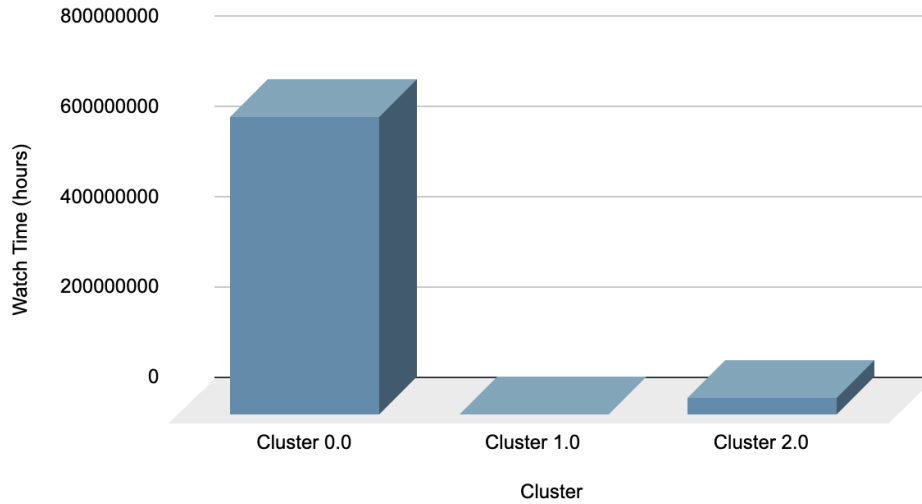


Figure 1: Total Potential Watch Time Volume by Cluster.

This 3D bar graph displays the total potential watch time volume for each cluster, with cluster 0 having disproportionately high watch time.

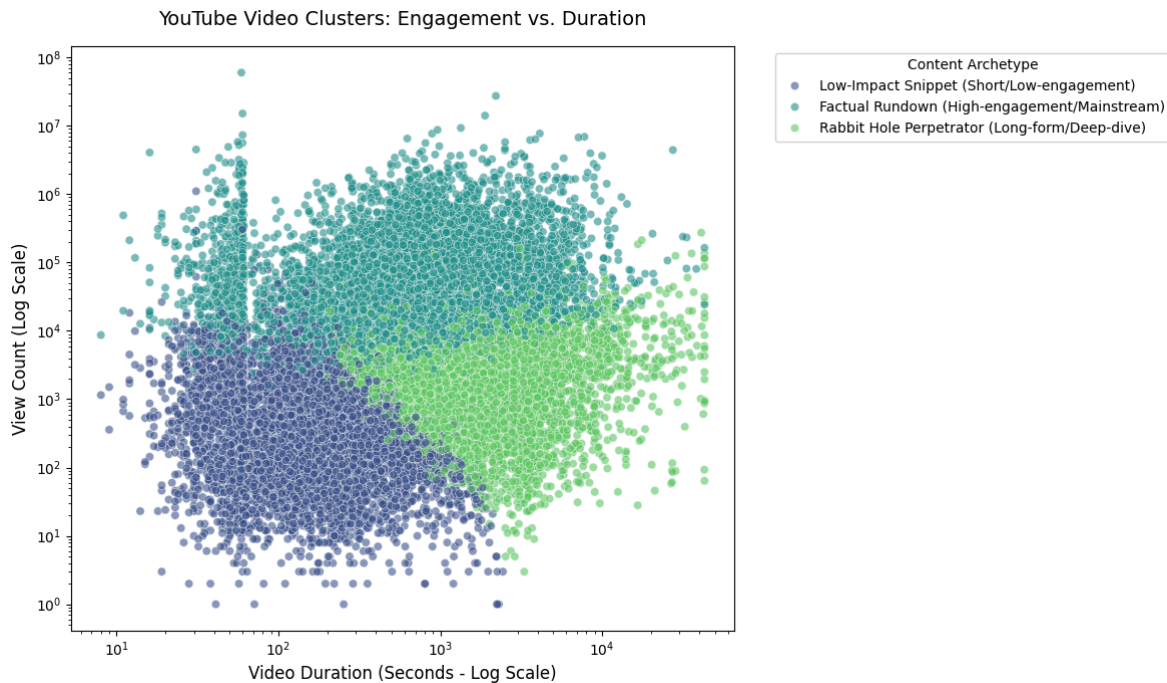


Figure 2: YouTube Video Clusters: Engagement vs. Duration

K-Means clustering of 22,193 videos. Maps the relationship between view count and video duration. Cluster 2 shows the emergence of the Rabbit Hole Perpetrator, long and with moderate engagement

cluster_name	views	likes	comments	duration_sec	video_count
Factual Rundown (High-engagement/Mainstream)	36737.5	1626.5	278.0	631.5	8060
Low-Impact Snippet (Short/Low-engagement)	370.0	6.0	1.0	157.0	8668
Rabbit Hole Perpetrator (Long-form/Deep-dive)	2081.0	65.0	9.0	1527.0	5465

Table 1

This table displays the profiles of the 3 clusters, including the median values of key engagement metrics, video lengths, and the number of videos in each cluster.

Discussion: Hannah

Overall, the results give us a clear idea of YouTube's content distribution structure. The recommendation algorithm dedicates a high-duration ecosystem for sensitive topics in politics and healthcare. Addressing our initial research question, we conclude that YouTube's tendency to prioritize watch time and high engagement limits users' exposure to balanced perspectives.

Our results showed that only a small fraction of viral videos account for the majority of watch time, and many videos receive little attention. This reveals the skewed nature of YouTube engagement data. High-engagement, informational content achieves peak attention through shorter, punchier durations. We could oversimplify this and only focus on the fact that the Factual Rundown videos capture the majority of attention. However, the Rabbit Hole videos are important to explore. While their audience is much smaller, they capture users' attention for extended periods, influencing the content they consume and, ultimately, their perspectives on important issues.

Addressing our research question regarding the algorithm's preference for watch time, we see that the Rabbit Hole cluster, while not nearly as high as Factual Rundown, still collects a significant watch time. The YouTube algorithm's preference for watch time explains why it rewards rabbit-hole videos. The Factual Rundown cluster contains the vast majority of overall watch time. However, the Rabbit Hole cluster's 5.5% share of total watch time is particularly concerning given its depth. The algorithm pushes mainstream snippets to massive audiences, but for the smaller audience that goes down the rabbit hole, 25-plus-minute videos offer a prolonged ideological immersion that mainstream content does not. The attention captured by these long, deep-dive videos means users are gaining prolonged exposure to specific perspectives. Thus, we explored what specific content made up the Rabbit Hole cluster; what do the main topics reveal?

We found these topics and concluded that the Rabbit Hole Perpetrator videos were thematically dense. The topics were not just generic and broad; rather, the content featured topics

commonly susceptible to misinformation and polarization. This trend makes sense, as the Rabbit Hole videos rely on rapid, intense cultural debates, identity-driven politics, and alternative discourse to sustain human attention. The Topic Entropy statistic also revealed that users watching long-form content are pulled further away from varied summaries and become trapped in dense ideological clusters, limiting their exposure to balanced perspectives. This confirms what we initially hoped to show: the recommendation pathway allows for polarization to persist among long-form videos to satisfy the algorithm's engagement goals. Considering how the YouTube algorithm works, this narrowing of content makes sense, as YouTube strings similar content together to optimize for watch time. Glimpsing into these rabbit holes and considering the topics, users are likely not just casually browsing but being submerged in concentrated, opinionated narratives.

This report specifically analyzed only political and healthcare content on YouTube, but the results have broader implications for social media in general. Algorithm-based platforms that treat engagement and watch time as key factors in their algorithms are likely to develop rabbit-hole patterns. Specifically, we identify TikTok as a potential primary driver of rabbit-hole structures.

Limitations and Future Work

One limitation of this study, as mentioned briefly in the Methodology, is that we did not map the specific network paths. We did have an alternative method—Topic Entropy—which still analyzes recommended content but does not dive into the trajectories of the “rabbit hole” path. We lack the nuance of tracking a user's journey, click by click. In the future, we could explore the network paths in more depth with more time and coding experience. We could then model the algorithmic drift and likely find valuable results.

We discovered another limitation in our method while examining our Natural Language Processing (NLP) approach applied to auto-generated spoken transcripts. It is important to note that, unlike formal documents, YouTube transcripts are highly conversational and unscripted. Our methodology strategically employs custom stopwords to remove common verbal fillers such as “uh,” “um,” and “like.” Still, our BERTopic model failed to remove persistent vlogger terms such as “guys,” “got,” and “today.” As a result, some of the clusters were obscured by meaningless noise from these words. By default, BERTopic places non-cohesive documents into the outlier bin (Topic -1), so this cluster had a large amount of this vocabulary. To address this limitation, we could develop a more advanced text-cleaning pipeline and implement Part-of-Speech tagging to remove as much unnecessary language as possible.

References

- [1] M. H. Ribeiro, R. Ottoni, R. West, V. A. F. Almeida, and W. Meira, “Auditing radicalization pathways on YouTube,” *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, pp. 131–141, Jan. 2020, doi: <https://doi.org/10.1145/3351095.3372879>

- [2] M. Ledwich and A. Zaitsev, “Algorithmic Extremism: Examining YouTube’s Rabbit Hole of Radicalization,” *arXiv.org*, Dec. 24, 2019. <https://arxiv.org/abs/1912.11211>
- [3] M. Haroon, M. Wojcieszak, A. Chhabra, X. Liu, P. Mohapatra, and Z. Shafiq, “Auditing YouTube’s recommendation system for ideologically congenial, extreme, and problematic recommendations,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 120, no. 50, Dec. 2023, doi: <https://doi.org/10.1073/pnas.2213020120>
- [4] M. Ledwich, A. Zaitsev, and A. Laukemper, “Radical Bubbles on YouTube? Revisiting Algorithmic Extremism with Personalised Recommendations,” *First Monday*, vol. 27, no. 12, Dec. 2022, doi: <https://doi.org/10.5210/fm.v27i12.12552>
- [5] B. Lewis, “Rabbit Hole,” *Routledge eBooks*, pp. 90–102, Oct. 2024, doi: <https://doi.org/10.4324/9781032674612-10>
- [6] D. O’Callaghan, D. Greene, M. Conway, J. Carthy, and P. Cunningham, “Down the (White) Rabbit Hole: The Extreme Right and Online Recommender Systems,” *Social Science Computer Review*, vol. 33, no. 4, pp. 459–478, Oct. 2015, doi: <https://doi.org/10.1177/0894439314555329>

Appendices

Shannon Entropy(H):

$$H = - \sum_{i=1}^k p_i * \log_2(p_i)$$

Topic	Count	Name	Representation
-1	185	-1_want_got_today_good	[want, got, today, good, time, think, really, ...
0	242	0_trump_president_biden_harris	[trump, president, biden, harris, country, sta...
1	199	1_oh_got_good_let	[oh, got, good, let, did, think, want, guys, r...
2	68	2_god_lord_jesus_pray	[god, lord, jesus, pray, spirit, healing, fath...

Topic	Count	Name	Representation
3	55	3_thank_health_today_board	[thank, health, today, board, community, menta...
4	50	4_health_kidney_really_food	[health, kidney, really, food, animals, eat, d...
5	47	5_god_jesus_lord_praise	[god, jesus, lord, praise, come, hallelujah, g...
6	39	6_life_feel_mind_brain	[life, feel, mind, brain, depression, think, p...
7	35	7_black_rolling_network_media	[black, rolling, network, media, applause, mar...
8	25	8_mr_court_state_case	[mr, court, state, case, honor, jury, defendan...
9	15	9_lgbtq_queer_plus_news	[lgbtq, queer, plus, news, community, gay, eve...

Table 2: Cleaned Topics Discovered in the Rabbit Hole Cluster
Using BERTopic modeling, we identified the topics in the Rabbit Hole Cluster.